

FIPS Experiment Configuration Reference

A practical guide to the variables in `fips_config.py`: variable name -> what it controls. Current values shown below match the latest one-hemifield build used for scanner/dimension testing.

Current scan design	Spatial layout	Attention task
10 runs 24 trials/run 6:00 per run	Fixation: 6 dva from screen center Stimulus center: 2 dva opposite side	4 px fixation dot red / black / white press on color change

Units: DVA = degrees of visual angle; PX = pixels; S = seconds. Values ending in `_DVA` are converted to pixels using the selected monitor geometry.

Important: `PREVIEW_MODE` and fake preview displacements are for geometry/testing only. For real data collection, keep `PREVIEW_MODE = False` and use the participant's prescan summary.

1. Debug, preview, and run defaults

Variable	Current value	What it does
<code>CODE_TEST</code>	False	Turns code-test mode on/off. When enabled, the main script limits the run to <code>CODE_TEST_N_TRIALS</code> .
<code>DEBUG_ON_VSCODE</code>	False	Debug flag retained by the experiment. Use only if your local development workflow depends on it.
<code>TASK</code>	"FIPS"	Experiment/task label used in filenames and bookkeeping. Normally leave as FIPS.
<code>PREVIEW_MODE</code>	False	Allows scan previewing without a real prescan displacement. Set True only for geometry/timing tests.
<code>PREVIEW_PRESCAN_LEFT_DVA</code>	0.30	Fake leftward perceptual-match displacement, in dva, used only when <code>PREVIEW_MODE</code> is True and no prescan CSV exists.
<code>PREVIEW_PRESCAN_RIGHT_DVA</code>	0.30	Fake rightward perceptual-match displacement, in dva, used only when <code>PREVIEW_MODE</code> is True and no prescan CSV exists.
<code>DEFAULT_SUB_ID</code>	"99"	Default subject ID shown by the experiment GUI.

Variable	Current value	What it does
DEFAULT_SUB_NAME	"PT"	Default subject initials/name shown by the experiment GUI.
DEFAULT_RUN	"1"	Default run number shown by the experiment GUI.
DATA_DIR	"../data/psycphys"	Folder, relative to fips_one_hemi.py, where behavioral data, logs, trial files, and related outputs are stored.
CODE_TEST_N_TRIALS	4	Number of trials retained when CODE_TEST = True.

2. Monitor and window geometry

These values determine the pixel-per-degree conversion. For scanner work, the physical viewing distance, screen width, and effective pixel resolution must match the actual setup.

Variable	Current value	What it does
MONITORS["..."]["mon_dist"]	varies	Viewing distance from participant's eyes to display, in centimeters.
MONITORS["..."]["size_cm"]	varies	Physical display size used for DVA conversion. The first value is screen width in cm.
MONITORS["..."]["size_px"]	varies	Display resolution in pixels used by PsychoPy for geometry calculations.
MONITORS["..."]["refresh_rate"]	60	Expected monitor refresh rate in Hz. Used for frame timing and motion calculations.
MONITORS["..."]["screen"]	0	PsychoPy display/screen index to open the experiment on.
DEFAULT_MONITOR	"Scanner"	Monitor profile selected by default in the GUI.
FULLSCREEN	False	Whether the PsychoPy stimulus window opens fullscreen.
ALLOW_GUI	False	Whether GUI/window-system controls are allowed on the PsychoPy stimulus window.
BACK_COLOR	[0.2, 0.2, 0.2]	Experiment background color in PsychoPy rgb space, where each channel runs from -1 to +1.

Profile	Distance	Width	Resolution	Refresh
Macbook	50 cm	42 cm	1440 x 900	60 Hz
iMac	60 cm	30 cm	1920 x 1080	60 Hz
TseLab	63 cm	47.2 cm	1920 x 1080	60 Hz
Beaver	80 cm	52.3 cm	1920 x 1080	60 Hz
Scanner	128.7 cm	42.8 cm	1920 x 1080	60 Hz

3. EyeLink and scanner

Variable	Current value	What it does
EYELINK_HOST	"100.1.1.1"	IP address used to connect to the EyeLink host computer.
EYELINK_CALIBRATION_TYPE	"HV9"	EyeLink calibration pattern. HV9 is a nine-point horizontal/vertical calibration.
EYELINK_CALIBRATION_TARGET_SIZE_PX	24	Size of the EyeLink calibration target in pixels.
DEFAULT_EYELINK_CONNECTED	"yes"	Default GUI choice for real EyeLink connection. Switch to no/dummy mode for laptop testing.
REQUIRE_FIXATION_BEFORE_TRIAL	False	If True, the experiment waits for gaze to fall inside the fixation gate before a trial can proceed. Current scan design leaves this off.
FIXATION_GATE_DURATION_S	0.2	How long gaze must stay within the fixation gate when REQUIRE_FIXATION_BEFORE_TRIAL is enabled.
FIXATION_GATE_RADIUS_DVA	1.5	Half-width/half-height of the permitted fixation region, in dva, for the optional trial-start gaze gate.
EYE_USED	1	Eye used by the fixation gate: 1 = right eye, 0 = left eye.
SCANNER_TRIGGER_KEY	"5"	Keyboard key interpreted as the scanner trigger. The run clock starts after this key is received.

4. One-hemifield spatial layout

Variable	Current value	What it does
HEMIFIELD	"right"	Which visual hemifield receives the FIPS stimulus. "right" puts the stimulus on the right and fixation on the left; "left" mirrors both horizontally.
FIXATION_ECCENTRICITY_DVA	6.0	Horizontal distance of the fixation point from screen center, in dva. Current value is 6 dva, not 8.
STIMULUS_ECCENTRICITY_DVA	2.0	Horizontal distance of the stimulus center from screen center, in dva, on the opposite side from fixation.
FIXATION_Y_DVA	0.0	Vertical position of the fixation point relative to screen center, in dva.
STIMULUS_Y_DVA	0.0	Vertical position of the FIPS stimulus center relative to screen center, in dva.

Current right-hemifield geometry: fixation at $x = -6$ dva, stimulus center at $x = +2$ dva. Therefore fixation-to-stimulus-center separation = 8 dva.

5. Fixation point and attention task

Variable	Current value	What it does
FIXATION_DIAMETER_PX	4	Diameter of the fixation dot in pixels. This matches the Kay-style retinotopy fixation size.
FIXATION_ALPHA	0.5	Fixation-dot opacity. 0 = transparent, 1 = fully opaque.
FIXATION_COLORS_RGB255	red / black / white	The RGB255 colors the fixation dot can take during the attention task.
FIXATION_COLOR_NAMES	["red","black","white"]	Text labels paired with FIXATION_COLORS_RGB255 and written to attention-task logs.
FIXATION_TASK_HZ	15.0	Effective timing grid for fixation-color changes. The current Kay-style rule is quantized to 1/15 s.
FIXATION_CHANGE_MEAN_S	3.0	Mean interval between fixation-color changes, in seconds.
FIXATION_CHANGE_PLUS_MINUS_S	2.0	Random range around the mean interval. Current settings yield approximately 1-5 s between changes.
FIXATION_RESPONSE_KEYS	["1","2","3","4"]	Keys accepted as attention-task button responses. Change this to match the scanner response box mapping if needed.

6. Probe and moving-frame dimensions

Variable	Current value	What it does
DOT_RADIUS_DVA	0.5	Radius of each probe in dva.
DOT_DIAMETER_DVA	1.0	Diameter of each probe in dva. Keep consistent with DOT_RADIUS_DVA if you change either.
DOT_VERTICAL_OFFSET_DVA	2.0	Vertical offset of each probe from the stimulus center. One probe is +2 dva and the other is -2 dva.
FRAME_WIDTH_DVA	8.0	Horizontal size of the moving square/frame in dva.
FRAME_HEIGHT_DVA	8.0	Vertical size of the moving square/frame in dva.
FRAME_LINEWIDTH_DVA	0.25	Thickness of the moving frame border in dva.

Variable	Current value	What it does
FRAME_COLOR	[0.3, 0.3, 0.3]	Moving-frame color in PsychoPy rgb space (-1 to +1 per channel).
FRAME_TRAVEL_DVA	7.0	Total horizontal distance the frame travels from one endpoint to the other, in dva.
STIMULUS_CHOICES	["checkerboard", "plain", "old"]	Allowed stimulus rendering modes shown by the GUI.
DEFAULT_STIMULUS	"checkerboard"	Stimulus rendering mode selected by default.

7. Prescan timing configuration

Variable	Current value	What it does
PRESCAN["n_trials"]	16	Number of psychophysical matching trials in the prescan.
PRESCAN["n_runs"]	1	Number of prescan runs.
PRESCAN["n_blocks"]	1	Number of logical blocks in the prescan.
PRESCAN["cycle"]	8	Number of complete alternating motion/flash cycles used to determine stimulus duration.
PRESCAN["pre_run_baseline"]	0.0 s	Fixation-only baseline before prescan trials.
PRESCAN["post_run_baseline"]	0.0 s	Fixation-only baseline after prescan trials.
PRESCAN["dot_duration"]	0.05 s	Duration of the endpoint probe flash within each half-cycle.
PRESCAN["reversal_time"]	0.45 s	Frame-motion duration before the 50 ms endpoint flash/reversal.
PRESCAN["ITI"]	0.002 s	Prescan inter-trial interval.

8. Scanner-run timing configuration

Variable	Current value	What it does
SCAN["n_trials"]	24	Number of trials in each scanner run.
SCAN["n_runs"]	10	Total number of FIPS scanner runs.
SCAN["n_blocks"]	3	Number of mini-blocks per run. Current schedule uses 8 trials per mini-block, with 2 trials from each condition.
SCAN["cycle"]	8	Number of complete motion/flash cycles per trial. With current 0.45 + 0.05 s half-cycles, this produces an 8 s stimulus.
SCAN["pre_run_baseline"]	15.0 s	Fixation-only baseline after the scanner trigger and before Trial 1.
SCAN["post_run_baseline"]	15.0 s	Fixation-only baseline after the final trial.
SCAN["dot_duration"]	0.05 s	Probe flash duration at each frame endpoint.
SCAN["reversal_time"]	0.45 s	Frame-motion duration before each endpoint probe flash/reversal.
SCAN["ITI"]	6.0 s	Fixation-only interval between scanner trials.

Current trial timing: 450 ms motion + 50 ms flash = 500 ms per half-cycle; 16 half-cycles = 8 s stimulus. With 24 trials, 23 ITIs, and 15 s pre/post baselines, the programmed run length is 360 s = 6:00.

9. Condition labels

Variable	Current value	What it does
COND_FIPS	"FIPS"	Label used for moving-frame + aligned flashing-probe trials.
COND_VERIDICAL	"Veridical"	Label used for aligned flashing probes with no moving frame.
COND_PERCEPTUAL	"Perceptual"	Label used for physically displaced probes matched to the participant's prescan perceptual displacement.
COND_FRAME_ONLY	"FrameOnly"	Label used for moving-frame trials with no probes.
SCAN_CONDITIONS	4 labels	Master list of scan-condition names. Normally keep this consistent with the condition-building logic in the main script.

10. Quick-change reference for scanner testing

If you want to...	Change...
Move fixation farther from center	FIXATION_ECCENTRICITY_DVA
Move FIPS stimulus farther from center	STIMULUS_ECCENTRICITY_DVA
Make moving square larger/smaller	FRAME_WIDTH_DVA / FRAME_HEIGHT_DVA
Change square travel distance	FRAME_TRAVEL_DVA
Change square border thickness	FRAME_LINEWIDTH_DVA
Move probes farther apart vertically	DOT_VERTICAL_OFFSET_DVA
Make probes larger/smaller	DOT_RADIUS_DVA / DOT_DIAMETER_DVA
Change fixation-dot size	FIXATION_DIAMETER_PX
Change fixation response keys	FIXATION_RESPONSE_KEYS
Preview without a prescan	PREVIEW_MODE
Change trial count / run count	SCAN["n_trials"] / SCAN["n_runs"]
Change ITI or baselines	SCAN["ITI"] / SCAN["pre_run_baseline"] / SCAN["post_run_baseline"]

Best practice: change values in fips_config.py, not in fips_one_hemi.py. For real scanner acquisition, verify the Scanner monitor geometry first, keep PREVIEW_MODE off, and use the real prescan displacement for Perceptual trials.